When Does Graph Structure Help Noisy-Label Intrusion Detection? A Leakage-Audited Study of Rare-Class Training-Signal Retention

Abstract

Noisy-label training pipelines can discard low-frequency samples that carry much of the learnable signal for rare classes. We study whether a multi-view graph consistency diagnostic and soft sample weighting retain that signal without treating benchmark rows as operational or forensic evidence. We extend CoLD with Graph-CDM, a label-space diagnostic over the graph views supported by each dataset, and compare soft weighting with a shared-pipeline hard-deletion ablation. Experiments use CICIDS-2017 and UNSW-NB15 for intrusion-detection classification; CESNET-TLS-Year22 is explicitly included as a long-tail TLS service-classification stress test, not as an attack dataset. Under the current leakage-audited protocol, Graph-CDM's aggregate Macro-F1 gain over CoLD is dataset dependent: it is largest on CICIDS-2017 and at most +0.01 on the other two datasets. Under 40–80% asymmetric training-label noise, soft weighting improves Tail-F1 over hard deletion by +0.073, +0.022, and +0.040 on CICIDS, CESNET, and UNSW, respectively; no statistically significant aggregate Macro-F1 difference was detected by the current analysis. Confident Learning outperforms Graph-CoLD on aggregate and tail metrics on all three datasets. A leakage audit further shows that exact duplicates (24% of CICIDS-2017 training rows) and clean-label paths can substantially inflate graph-denoising results. The

supported claim is therefore mechanism-level and bounded: soft retention can preserve rare-class training signal within this synthetic-noise benchmark pipeline, but it is neither a universal accuracy improvement nor evidence of SOC or forensic effectiveness.

Keywords: intrusion detection, noisy labels, graph learning, training-signal retention, leakage audit, rare-class robustness

1 Introduction

Machine-learning classifiers underpin modern network intrusion-detection systems (IDS) and security operations centers (SOCs). These models are trained from labels that are seldom perfect: they may be delayed, produced by rules of uneven quality, or shaped by analyst workflows that emphasize high-severity cases [2, 3]. The resulting label noise degrades both detection quality and operational triage. Crucially, the cost of label noise is asymmetric. Discarding a mislabeled benign flow may raise a benchmark score, but discarding an informative low-frequency sample can remove much of the training signal available for a rare class.

This asymmetry motivates a reframing of the problem. Conventional noisy-label pipelines treat suspicious samples as contamination to be purged; we instead treat data curation as a *training-signal retention* problem, in which the decision to delete weighs estimated label unreliability against the class-level cost of losing scarce training examples. This is a mechanism-level learning claim: benchmark rows are not, by themselves, SOC investigation or forensic evidence. The question is therefore not only *how accurately* noise can be detected, but *what should be done* with samples the detector cannot confidently classify: delete, trust, or retain at reduced influence.

CoLD [1] recently framed noisy-label IDS through *local consistency*: because different traffic classes can share feature neighborhoods, label noise induces spurious

associations, and a collaborative representation can identify and remove likely-noisy samples before training. CoLD, like most dataset-purification methods, ultimately performs *hard deletion*. We take CoLD as our baseline and ask two questions. (Q1) Does lifting the diagnostic from independent samples to a multi-view graph, kept strictly in label space, improve robustness? (Q2) Does replacing hard deletion with signal-preserving soft weighting—which down-weights rather than deletes suspicious-but-informative samples—retain rare-class training signal and improve tail-class detection?

Answering these questions honestly required an unusual amount of methodological hygiene. Our first implementations produced spectacular gains that did not survive scrutiny: a leakage audit revealed that (i) CICIDS-2017 contains 24% exact-duplicate records that let neighborhood consistency trivially recover labels, and (ii) any use of clean-label signal in the scoring or graph construction inflates results. After removing both, the apparent advantage shrank to a modest, defensible level, and the headline metric we had used for signal retention turned out to be tautological. We report this process because it is, we believe, a contribution in its own right: graph-based denoising results in security are easy to inflate and must be audited.

Contributions.

The current empirical record supports the following bounded contributions:

1. A *leakage audit* that quantifies duplicate- and clean-label-path inflation and documents how apparently large graph-denoising gains can collapse under a de-oracled protocol (Section 6.4).
2. A *signal-preserving soft weighting* rule that improves Tail-F1 over hard deletion under high asymmetric noise within an otherwise shared pipeline (Section 6.3). The result concerns rare attack classes on CICIDS/UNSW and low-frequency service

classes on CESNET; the current aggregate analysis detects no significant difference but does not establish equivalence.

3. Graph-CDM, a *label-space* multi-view consistency diagnostic used to study when graph structure helps. Its aggregate gain over CoLD is modest and dataset dependent, and the three-dataset informativeness pattern is descriptive, not inferential (Sections 6.1 and 6.5).

We deliberately bound our claims. Graph-CoLD is not a universal accuracy improvement, and retained benchmark samples are not operational evidence. Its distinctive mechanism-level value is retaining rare-class training signal that hard deletion can discard.

2 Related Work

Noisy-label learning.

Approaches divide into robust-training and dataset-purification families. Robust training modifies losses or training dynamics: loss correction with a transition matrix [4], peer-network small-loss selection (Co-teaching [5], Co-teaching+ [6]), disagreement-based updates (Decoupling [7]), and semi-supervised treatment of suspected-noisy data (DivideMix [8]). Purification detects and removes mislabeled instances: confident learning [9], representation-based filtering (FINE [14]), and, in security, MORSE [15] and MCRe [16]. CoLD [1] is the closest prior work and our baseline. Our departure is twofold: we keep the diagnostic in label space (avoiding the distance-metric fragility CoLD itself criticizes), and we replace deletion with signal-preserving weighting motivated by the semi-supervised insight that suspected-noisy samples carry usable signal [8, 15].

Graph learning for security telemetry.

Security events are relational, and graph neural networks [17–20] naturally propagate contextual signals. Provenance-graph IDS such as Flash [22] and Argus [23] demonstrate the value of structure for enterprise detection. We use graph representation learning as a backbone but, unlike embedding-distance denoisers, keep the noise diagnostic in label space so it stays aligned with the classifier.

SOC alert handling and concept drift.

Operational systems must also cope with alert volume and distribution shift. Alert-classification and triage work [25, 26] and drift-aware detection [24] motivate our operational framing, though our contribution is denoising for rare-class signal retention rather than a deployed triage study.

Contrastive self-supervision.

Our representation stage uses cross-view contrastive learning [21], adapted so that the same node in two views forms a positive pair.

2.1 Positioning and Open Challenges

Noisy-label learning for intrusion detection

Learning from noisy labels has become an important challenge in security analytics because network telemetry is often collected through heterogeneous sources, automated rules, and delayed analyst feedback. Unlike conventional computer vision benchmarks where label corruption can often be controlled synthetically, security labels may reflect incomplete attack knowledge, imbalanced detection rules, and evolving adversarial behaviors [30, 31]. Therefore, improving robustness requires not only higher classification accuracy but also understanding which samples should be trusted and which samples contain useful but uncertain training signal.

Existing noisy-label learning methods can be broadly divided into robust training and dataset purification. Robust training approaches attempt to reduce the influence of corrupted samples during optimization. Loss correction methods estimate label transition characteristics and modify the training objective [32]. Sample-selection approaches exploit uncertainty, loss dynamics, or disagreement between models to identify potentially noisy instances [38, 39]. Co-teaching and related disagreement-based approaches train multiple networks and exchange selected small-loss samples to reduce memorization of corrupted labels [5, 6]. Decoupling further separates the decision of when to update from how to update, demonstrating that disagreement between classifiers can provide useful information under label corruption [7].

A different family focuses on dataset purification. Confident Learning estimates label uncertainty by identifying statistically inconsistent annotations [9]. FINE explores representation-space filtering of noisy instances through latent feature structures [14]. In security domains, MCRe and MORSE extend noisy-label handling toward malicious traffic and malware classification scenarios [15, 16]. More recently, intrusion detection studies have explored contrastive learning and representation enhancement as a way to improve robustness against noisy or incomplete observations [33, 34].

CoLD provides the closest foundation for our work by introducing collaborative label denoising specifically for network intrusion detection. However, existing purification-oriented methods generally treat suspicious samples as removable noise. This assumption is problematic for imbalanced classification because rare-class samples may simultaneously appear inconsistent and be important to the learned decision boundary. Graph-CoLD therefore does not aim to replace purification, but instead explores whether structural context and signal-preserving weighting can retain informative suspicious instances.

Graph learning and contrastive representation for security

Graph representation learning has become increasingly important for security analytics because network activities naturally form relational structures among hosts, addresses, sessions, and temporal events. Graph convolutional networks and graph attention networks demonstrated that neighborhood information can improve representation learning when samples are connected through meaningful relations [17, 18]. Inductive graph representation learning further enables scalable modeling of previously unseen nodes [19], while heterogeneous graph transformers extend graph reasoning to multiple relation types [20].

In security applications, graph methods have mainly focused on attack detection, provenance analysis, and contextual behavior modeling. Systems such as Flash demonstrate the value of provenance graph representation learning for enterprise intrusion detection [22]. Other work investigates security-oriented representation learning and graph analytics for detecting suspicious behaviors [23]. These approaches usually focus on discovering malicious structures from graph embeddings.

Graph-CoLD differs from these methods in its objective. The graph structure is not directly used as an anomaly detector; instead, it provides contextual signals for a label-space consistency diagnostic. This distinction is important because embedding distance alone may capture feature similarity while remaining disconnected from the classifier's final prediction behavior. By combining graph context with prediction-space disagreement, Graph-CoLD aims to identify unreliable labels while preserving samples that remain informative.

SOC alert triage, prioritization, and concept drift

Modern SOC environments face not only detection accuracy challenges but also large-scale alert management problems. Excessive alerts, evolving attack behaviors, and

changing network distributions create operational pressure for analysts. Alert classification and active-learning approaches have explored ways to reduce analyst workload by improving alert categorization [25]. Reinforcement-learning-based alert prioritization methods further investigate how automated systems can rank alerts according to operational importance [26].

Beyond static classification, concept drift has become a critical challenge in long-running security monitoring. Detection systems must adapt when normal traffic patterns, attack strategies, and service behaviors evolve. Studies such as CADE, RecDA, and Insomnia investigate drift detection and adaptation mechanisms for network intrusion detection [24, 40, 41]. Recent SOC-oriented research also explores alert aggregation and filtering through deep learning models to improve investigation efficiency [35–37].

These operational studies motivate our security framing, but Graph-CoLD addresses a different layer of the problem. Rather than replacing analyst triage or predicting attack priority directly, we focus on improving the quality of the training data that supports downstream detection. The key difference is that Graph-CoLD combines three principles: label-space consistency diagnosis, signal-preserving retention, and leakage-audited evaluation. This design aims to answer when graph-based denoising genuinely helps security analytics and when structural information is insufficient.

3 Problem Formulation

Let V be the set of training flows and $\mathcal{M}$ the set of *active* graph views. Each view $m \in \mathcal{M}$ induces a graph $G^m = (V, E^m)$ over the same nodes. The observed training label $\tilde{y}_v$ may differ from the unobserved clean label y_v. We seek a classifier that (i) resists the effect of noisy labels, and (ii) retains useful training signal—especially clean samples of rare classes that a purifier is tempted to discard.

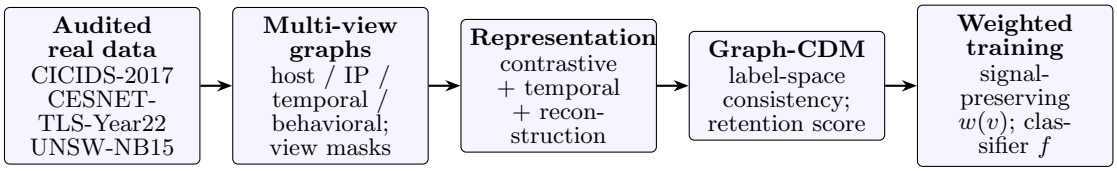


Fig. 1 Graph-CoLD pipeline. Only views supported by a dataset's verified schema are activated; unsupported views are disabled rather than fabricated. The consistency diagnostic (Graph-CDM) is computed in label space and feeds a signal-preserving soft weight and a downstream classifier. Datasets are CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 (active views vary by dataset; see Table 1).

We adopt three design principles. First, the noise diagnostic operates in *label/prediction space*, not embedding distance, so it remains aligned with the classifier and does not re-introduce the local-consistency fragility of distance-based purifiers. Second, denoising is *signal preserving*: a suspicious but informative sample is down-weighted rather than removed. Third, the evaluation is traceable to audited real data, with explicit scope limits.

4 Method

Graph-CoLD is a two-stage extension of CoLD: self-supervised multi-view representation learning, followed by a label-space consistency diagnostic that drives a signal-preserving weight (Figure 1).

4.1 Multi-view graph construction

Views are built from semantically related feature blocks over the training nodes: a *host* view links samples sharing endpoint identifiers; an *IP* view uses communication/flow statistics, ports, protocol and TLS metadata; a *temporal* view links samples within a time window; a *behavioral* view groups service, protocol, port-state, and related behavioral feature blocks (it is a feature-block view, not host-process lineage) and is used only where such fields exist. Support is part of the reproducible dataset contract: CICIDS-2017 activates host/IP/temporal, CESNET-TLS-Year22 activates IP/temporal, and UNSW-NB15 activates temporal/behavioral. Threat-intelligence views are disabled

when the fields are absent. Crucially, edges never cross the train/test split, and (Section 6.4) exact/near-duplicate edges are removed.

4.2 Self-supervised representation

Each active view produces a node embedding z_v^m from Bernoulli-masked input features, $x'_v = x_v \odot \text{Bernoulli}(1-\delta)$. Cross-view contrastive learning treats the same node in two views as a positive pair and other batch nodes as negatives:

$$\mathcal{L}_{\text{con}} = -\log \frac{\exp(\text{sim}(z_v^a, z_v^b)/\tau)}{\sum_u \exp(\text{sim}(z_v, z_u)/\tau)}. \tag{1}$$

Temporal alignment and reconstruction terms regularize the embedding, $\mathcal{L}_{\text{rep}} = \mathcal{L}_{\text{con}} + \alpha\mathcal{L}_{\text{temporal}} + \beta\mathcal{L}_{\text{recon}}$; active views are fused by mean aggregation to obtain z_v.

4.3 Graph-CDM: a label-space consistency diagnostic

For each node v,

$$\text{GraphCDM}(v) = \lambda_1 D_{\text{pred}}(v) + \lambda_2 D_{\text{neigh}}(v) + \lambda_3 D_{\text{view}}(v), \tag{2}$$

where the prediction term compares per-view predicted labels with the observed *noisy* training label,

$$D_{\text{pred}}(v) = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}[\tilde{y}_v^m \neq \tilde{y}_v], \tag{3}$$

the neighborhood term is a label-space KL divergence between a node's soft prediction and its neighborhood mean, $D_{\text{neigh}}(v) = \text{KL}\big(\hat{y}_v \,\|\, |N(v)|^{-1}\sum_{u\in N(v)} \hat{y}_u\big)$, and the view term measures active-view mode disagreement, $D_{\text{view}}(v) = 1 - \max_c \frac{1}{M}\sum_m \mathbf{1}[\tilde{y}_v^m = c]$. A fourth provenance-chain term explored in early drafts is removed (frozen λ_4=0): none of the three datasets carries verified provenance chains, and no reported result

depends on it. No term uses clean labels or the noise mask; this is enforced by a regression test.

4.4 Signal-preserving soft weighting

The internal evidence score protects samples that would be costly to discard, $e(v) = \text{freq_protect}(\tilde{y}_v)\,(1 + \gamma\,\text{anom}(v))$, min-max normalized to $\tilde{e}(v)$. Rather than a small additive floor (which collapses to hard deletion), we use an evidence-*rescue* form so that suspicious yet high-evidence samples retain real weight:

$$w(v) = \underbrace{\sigma\big(-\kappa(\text{GraphCDM}(v) - \theta)\big)}_{\text{clean branch}} + \underbrace{\big(1 - \sigma(\cdot)\big)\,\lambda_r\,\tilde{e}(v)}_{\text{evidence rescue}}. \tag{4}$$

The classifier minimizes $\sum_v w(v)\,\text{CE}(f(z_v), \tilde{y}_v)$. Setting $\lambda_r = 0$ and thresholding recovers hard deletion; this is our `ablation_hard` comparator, sharing the identical graph, representation, and diagnostic so that the ablation isolates *only* the weighting. Two further frozen comparators isolate whether the specific *rescue form* (rather than retention per se) carries the benefit: a frequency-floor reweighting $w(v) = \max(\tau_{\text{ret}}, \text{freq}(v))$ with $\text{freq}(v) = \min(1, \text{median class count}/\text{count}(\tilde{y}_v))$, and a no-graph class-balanced cross entropy with effective-number weights ($w_c \propto (1-\beta)/(1-\beta^{n_c})$, β=0.9999) [10]; both share the encoder and training budget (pre-registered family C, Section 6.3).

We stress the intended interpretation of this internal score: "evidence" here means a heuristic estimate of training-signal retention value, not SOC or forensic evidence. Frequency protection encodes *deletion cost*, not trustworthiness. A rare sample is protected because discarding it is costly—it may be one of few training examples for a tail class—not because rarity implies a clean label. The score is thus a heuristic proxy for a risk-aware retention signal $R_{\text{delete}}(v) \propto \text{Cost}(\tilde{y}_v)\,\big(1+\gamma\,\text{anom}(v)\big)$, where Cost is high for rare classes, and the weighting in Eq. (4) implements risk-aware sample retention:

samples with high diagnostic suspicion but high deletion cost keep reduced—never zero—influence. A full decision-theoretic treatment that models $P(\text{clean} \mid x) \cdot \text{Cost}(y)$ explicitly is left to future work; the empirical ordering in Section 6.3, where confident-learning-precision pruning outperforms our heuristic, indicates the value of sharpening the $P(\text{clean} \mid x)$ component. A structural limitation is that freq_protect is keyed on the *observed* noisy label $\tilde{y}_v$: under asymmetric tail-to-majority noise, a flipped tail sample is observed as the majority class and receives weaker protection, while a majority sample mislabelled into a tail class is over-protected. Consistent with this, the pre-registered family C comparison (Section 6.3) finds no detectable advantage of the rescue form over the simple frequency floor—retention itself, not the weighting form, carries the benefit. Both the CoLD-aligned proxy baseline and the `ablation_hard` variant use the same shared encoder architecture, training budget, and view-masking configuration as Graph-CoLD; they differ only in the presence or absence of graph views and the weighting rule applied after Graph-CDM scoring.

4.5 Triage-oriented benchmark proxy

A priority score $P(v) = \alpha_1 \hat{y}_{\text{mal}}(v) + \alpha_2 \text{GraphCDM}(v) + \alpha_3 \tilde{e}(v)$ models a triage-oriented ranking. We evaluate it only by benchmark labels and, as we show, Graph-CoLD ties strong peers on top-K precision; without an analyst study, we claim neither a ranking advantage nor operational effectiveness.

4.6 Graph-CoLD Optimization Procedure

The complete Graph-CoLD pipeline consists of representation learning, label-space consistency estimation, retention scoring, and weighted classifier optimization. Algorithm 1 summarizes the complete procedure.

Algorithm 1: Graph-CoLD signal-preserving noisy-label learning.

Input: Multi-view graphs $\{G_m\}_{m=1}^{M}$, noisy labels $\tilde{Y}$, feature matrix X.

Output: Trained classifier f.

1. Construct active graph views according to available telemetry fields.
2. Learn view-specific representations using contrastive, temporal-alignment, and reconstruction objectives, then fuse them into node embeddings Z.
3. Generate prediction-space and neighborhood consistency statistics and compute a Graph-CDM score for each training sample.
4. Estimate internal retention scores from label frequency and anomaly factors, then compute the signal-preserving weights $w(v)$.
5. Optimize the classifier using weighted cross entropy and return f.

The design follows three considerations. First, the diagnostic operates in label and prediction space rather than pure embedding distance. This maintains consistency with the classifier decision process and avoids relying solely on representation similarity. Second, the weighting mechanism adopts an signal-rescue formulation because suspicious samples may still contain useful class information. A simple lower bound would approximate hard deletion, whereas the rescue formulation allows uncertain but informative samples to participate in training. Third, hard deletion is retained as the primary ablation because it isolates the contribution of signal preservation while keeping the same graph representation and diagnostic components.

4.7 Frozen configuration and tuning discipline

All experiments use one frozen configuration, fixed before the final experiment grid and archived with the reproduction package: diagnostic weights λ_{pred}=0.4, λ_{neigh}=0.3, λ_{view}=0.3 (λ_4=0, removed); weighting κ=4.0, θ=0.5, λ_r=0.8, γ=1.0, log-scaled freq_protect, retention floor τ_{ret}=0.1; hard-deletion threshold CDM $\leq$ 0.5 on the shared diagnostic. Graph construction: host/IP views use k=10 Euclidean nearest neighbours, the temporal view links k=2 neighbours within a 3,600 s window; the

encoder is a 2-layer, 4-head heterogeneous graph transformer (hidden 128, dropout 0.2), trained 100 epochs (stage 1) plus 50 (stage 2), learning rate 10^{-3}, batch 128. *Tuning discipline:* no clean validation set and no true injected noise rate is available to any method at any point; hyperparameter selection uses defaults or criteria computed on the noisy training set only (e.g. cleanlab-estimated noise rates for the Co-Teaching forget schedule). This discipline is part of the frozen protocol and applies identically to every comparator.

5 Experimental Design

5.1 Datasets and protocol

Table 1 summarizes three benchmark datasets with different tasks. CICIDS-2017 and UNSW-NB15 are intrusion-detection datasets. CESNET-TLS-Year22 is a TLS service-classification dataset whose SNI-derived application labels do not carry attack/benign semantics; we retain it only as a long-tail service-class stress test and do not use it to support attack-level claims. CICIDS-2017 is used under an 11-class *post-filter* policy with exact deduplication (24% of raw training rows removed; Section 6.4). CESNET-TLS-Year22 is a year-spanning backbone TLS dataset used under a 25-service-class audit-window policy (it replaces MALTLS-22, for which we lacked a verified license path). UNSW-NB15 is used under its partition with the smallest class (Worms) removed. Noise semantics per dataset: the asymmetric flip target is benign for the two IDS datasets and a high-frequency service class for CESNET; legacy result filenames may use `tail-to-benign` as an implementation label, but that term has no semantic meaning for CESNET.

Frozen primary protocol.

All headline results use a *frozen final protocol*, fixed before the final experiment grid; any later deviation is disclosed as a protocol amendment (Appendix). The protocol has

Table 1 Datasets and active graph views. Source: `tables/table_1_dataset_protocol.csv`.

Dataset	Classes	Sample policy	Active views
CICIDS-2017	11	post-filter, exact-dedup	host/IP/temporal
CESNET-TLS-Year22	25	audit window	IP/temporal
UNSW-NB15	9	partition (no Worms)	temporal/behavioral

two ingredients per dataset. (i) *Proxy-field masking*: features that participate in each dataset's label construction, or are strong proxies of it, are removed before any graph or model sees the data (CICIDS: tier-1 mask; CESNET and UNSW: tier-2, which additionally removes the `CATEGORY` one-hot columns—parents of the CESNET label—and the `ct_*` generated features of UNSW). (ii) *Strongest-available group-disjoint split*: CICIDS uses a temporal split ordered by source-file day (with exact training-row deduplication); CESNET uses a strict temporal boundary with zero train–test entity overlap (S1∩S2); UNSW uses its official fixed partition, whose released files lack timestamp and entity columns (a limitation we measure rather than hide: the official partition itself contains 38.4% exact train–test feature collisions after masking).

Split taxonomy and audit measurements.

The audit distinguishes S0 (iid stratified), S1 (temporal), S2 (entity-disjoint, grouped so that no entity appears on both sides), and S3 (leave-one-scenario-out, defined only where the scenario is not identical to the class label: attack-day holdout for CICIDS; undefined for UNSW and CESNET, where S1∩S2 substitutes). For every split we measure and report the train–test exact and near-duplicate intersection (cosine similarity ≥ 0.999 on standardized features). Test-time inference is inductive (per-sample); a transductive variant that builds graphs over test batches is reported only as a control and does not represent an online IDS. On CICIDS, entity isolation is infeasible by construction—the flow graph collapses into one giant connected component spanning 99.96% of valid flows—which we document as a dataset-level finding (Section 9).

Noise injection.

Noise is injected into training labels only; test labels stay clean. The rare-class analysis uses asymmetric noise concentrated on tail classes (training class count $\leq$ window median) at rates $\{40, 60, 80\}\%$ with seeds 0–9 bound to the injection. Flip targets are the benign class for the two IDS datasets and the high-frequency service class `microsoft-diagnostic` for CESNET, which has no benign semantics. Symmetric and graph-consistency noise are reported as supplements. The complete mapping matrix ships with the reproduction package (`noise_map.json`).

Pre-registered statistical protocol.

The inference unit is the per-noise-rate paired seed comparison (n=10); we do not pool the 30 seed–rate observations as independent. Holm correction is applied within three pre-registered families: family A (9 tests; soft retention vs. hard deletion, Tail-F1), family B (18 tests; CL+Graph-CDM vs. CL, Macro- and Tail-F1—the G2 decision gate), and family C (18 tests; soft weight vs. simple reweighting baselines, Tail-F1). "No aggregate cost" claims are replaced by TOST non-inferiority at δ=0.01 Macro-F1; because a 0.01 macro change could hide a concentrated single-class regression, a per-class guard additionally requires no class-level F1 regression above 0.05. Seed random-intercept mixed-effects models provide robustness confirmation. Success requires Holm $p < 0.05$, $d_z \geq 0.3$, and a bootstrap 95% CI excluding zero; the verdict vocabulary is frozen (*robust*: $\geq$2 datasets pass; *narrow*: 1; *null*: 0).

Windows and scale.

Each dataset uses a deterministic audit window of 10,000 training and 5,000 test instances drawn inside the frozen split. A one-shot scale validation on CICIDS (train 100,000 / test 20,000; 3 seeds $\times$ 3 rates) checks that conclusions do not flip with scale; they do not (Section 6.3). We are explicit that this is a focused audit study, not a full-archive evaluation.

5.2 Comparators and metrics

Within the frozen protocol we compare six configurations that share one encoder architecture and training budget: Graph-CoLD-soft; hard deletion (`ablation_hard`); the frequency-floor and class-balanced-CE comparators of Section 4; Confident Learning (cleanlab `find_label_issues` with *full stratified out-of-fold* probabilities, per its sample-out protocol, and a shared-budget classifier head) [9]; and the CL+Graph-CDM combination that soft-reweights CL-retained samples (issue-set intersection/union fusions are exploratory). The rows labelled *CoLD-aligned proxy* were called "CoLD" in an earlier draft; the implementation predates the official CoLD code release and does not implement its multi-subset encoding, clustering-consistency purification, or XGBoost head. We therefore make no claim of reproducing or outperforming official CoLD, whose published numbers arise under a different nine-class protocol (Section 7.1). FINE [14] and a fully-implemented Co-Teaching [5] (forget schedule driven by an *estimated*, not oracle, noise rate) were run under the legacy stratified audit window and are reported separately as explicitly non-comparable (Section 6.3). Metrics: aggregate Macro-F1, tail-class Macro-F1 (tail = classes with training count $\leq$ window median, fixed per data contract), tail recall, and paired deltas with bootstrap 95% CIs under the statistical protocol above.

6 Results

6.1 RQ1: A label-space graph diagnostic modestly improves accuracy over CoLD

Table 2 reports per-dataset means. Graph-CoLD improves aggregate Macro-F1 over CoLD on all three datasets, with the gain concentrated on CICIDS, where host/IP/temporal views are most informative, and small on CESNET/UNSW. The improvements over CoLD are Holm-significant per dataset (CICIDS p =

Table 2 Per-dataset mean±std under high asymmetric tail noise (rates 40/60/80%, seeds 0–9; 30 scenarios per dataset). Sources: `tables/table_p2f_summary.csv` (Graph-CoLD family) and `results/p5_external_baselines.csv` (external anchors, same protocol). Best value per dataset and metric in bold.

Dataset	Method	Macro-F1	Tail-F1	Tail-recall
CICIDS-2017	CoLD	0.534±0.039	0.047±0.070	0.043±0.065
	ablation_hard	0.688±0.087	0.361±0.193	0.277±0.173
	Graph-CoLD-soft	0.721±0.086	0.434±0.191	0.328±0.180
	Graph-CoLD-semisup	0.687±0.093	0.357±0.206	0.272±0.181
	Confident-Learning	**0.769±0.048**	**0.605±0.103**	**0.517±0.112**
	Co-Teaching	0.567±0.071	0.118±0.152	0.128±0.171
CESNET-TLS-Year22	CoLD	0.658±0.130	0.411±0.247	0.306±0.211
	ablation_hard	0.680±0.132	0.453±0.249	0.331±0.218
	Graph-CoLD-soft	0.692±0.120	0.475±0.227	0.344±0.204
	Graph-CoLD-semisup	0.679±0.129	0.450±0.243	0.327±0.212
	Confident-Learning	**0.805±0.076**	**0.708±0.136**	**0.628±0.159**
	Co-Teaching	0.488±0.152	0.179±0.258	0.181±0.262
UNSW-NB15	CoLD	0.456±0.032	0.096±0.068	0.187±0.146
	ablation_hard	0.471±0.035	0.132±0.077	0.205±0.128
	Graph-CoLD-soft	0.485±0.033	0.173±0.078	0.229±0.120
	Graph-CoLD-semisup	0.469±0.033	0.130±0.074	0.194±0.119
	Confident-Learning	**0.499±0.026**	**0.254±0.085**	**0.317±0.123**
	Co-Teaching	0.419±0.036	0.055±0.080	0.120±0.178

1.0×10^{-3}, CESNET $p = 4.1 \times 10^{-2}$, UNSW $p = 4.1 \times 10^{-2}$; source `tables/table_p2d_per_dataset_vs_cold.csv`). Notably, `ablation_hard` already captures most of this gain: the graph and the label-space diagnostic, not the weighting, drive aggregate accuracy.

6.2 Performance under Different Noise Types

To better understand when Graph-CoLD provides benefits, we analyze performance across different label-noise mechanisms. Table 3 summarizes Macro-F1 results under clean, symmetric, asymmetric, and graph-consistency noise.

The results show that the benefit of Graph-CoLD is strongly dataset-dependent. On CICIDS-2017, where host, IP, and temporal relations provide richer structural signals, Graph-CoLD achieves consistent improvements over CoLD across all noise types.

Table 3 Graph-CoLD vs. CoLD Macro-F1 by noise type (mean over noise rates, seeds 0–9). Source: `tables/table_p4_noise_type_breakdown.csv`.

Dataset	Noise type	CoLD	Graph-CoLD-soft	Δ
CICIDS-2017	clean	0.671150	0.778689	0.107539
CICIDS-2017	symmetric	0.553919	0.592429	0.038510
CICIDS-2017	asymmetric	0.322779	0.429941	0.107162
CICIDS-2017	`graph_consistency`	0.496244	0.531555	0.035311
CESNET-TLS-Year22	clean	0.966321	0.968261	0.001941
CESNET-TLS-Year22	symmetric	0.898812	0.902585	0.003774
CESNET-TLS-Year22	asymmetric	0.698745	0.708302	0.009557
CESNET-TLS-Year22	`graph_consistency`	0.846591	0.849355	0.002764
UNSW-NB15	clean	0.494282	0.501909	0.007627
UNSW-NB15	symmetric	0.445324	0.445923	0.000599
UNSW-NB15	asymmetric	0.296512	0.303933	0.007421
UNSW-NB15	`graph_consistency`	0.434276	0.436233	0.001957

Under asymmetric noise, the improvement is 0.107 in Macro-F1. This observation is consistent with the hypothesis that graph context is more valuable when corruption is associated with non-uniform class behavior.

For CESNET-TLS-Year22 [42], the improvement is smaller because the baseline already achieves high performance under the evaluated protocol. Under this near-ceiling condition, additional graph information provides only marginal gains. UNSW-NB15 [43] shows a similar trend, indicating that graph denoising effectiveness depends on whether available views contain sufficient discriminative structure.

Overall, the noise-type analysis supports a bounded conclusion: Graph-CoLD is not universally superior under all conditions, but provides the largest benefit when the available relational views contain informative structure and when label corruption disrupts class-consistent training signal.

6.3 RQ2: Signal-preserving weighting helps tail classes, not aggregate accuracy

The central finding concerns *where* signal-preserving weighting matters. On aggregate Macro-F1, the current analysis detects no statistically significant difference

Table 4 Powered tail-class comparison: Graph-CoLD-soft vs. hard deletion under asymmetric tail noise ($n = 30$ scenarios/dataset). "No" denotes that no significant aggregate regression was detected; it is not an equivalence result. Source: `tables/table_p2f_powered_tests.csv`.

Dataset	ΔTail-F1	Holm p	Cohen's d_z	Sig. agg. reg.?
CICIDS-2017	+0.0731	1.5×10^{-7}	1.28	no
CESNET-TLS-Year22	+0.0218	3.7×10^{-5}	0.91	no
UNSW-NB15	+0.0405	9.6×10^{-8}	1.29	no

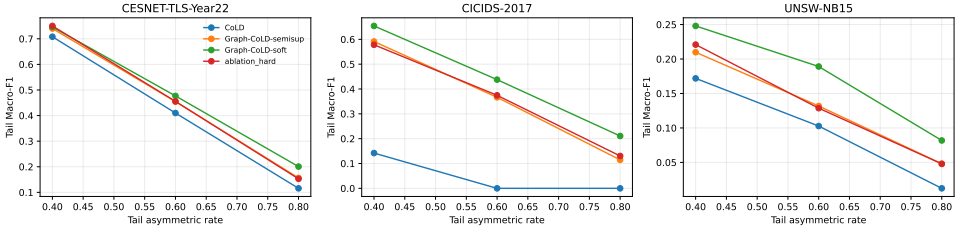


Fig. 2 Tail-class Macro-F1 of signal-preserving soft weighting vs. hard deletion across asymmetric noise rates. The gain is small but consistent and significant on all three datasets.

between soft weighting and hard deletion; this absence of significance does not establish equivalence. On tail classes, soft weighting significantly outperforms hard deletion (Table 4): ΔTail-F1 $= +0.073, +0.022, +0.040$ on CICIDS/CESNET/UNSW, all Holm-significant at $p < 10^{-4}$ with large effect sizes ($d_z = 1.28, 0.91, 1.29$), with no significant aggregate-Macro-F1 regression detected. For CICIDS and UNSW these tail classes are attack categories; for CESNET they are low-frequency service classes. Figure 2 shows the effect is stable across noise rates. The mechanism-level interpretation is that hard deletion can remove clean rare-class training examples flagged as suspicious, whereas soft weighting keeps them at reduced influence.

We stress an honesty point. An earlier "rare-evidence recovery rate" that counted a sample as recovered whenever it was *retained* was tautological. We redefine recovery to require correct downstream classification; hard deletion is then no longer automatically zero. Because soft retention still scores near-ceiling on this diagnostic,[1] we rely solely on test-set tail-F1 for all headline comparisons.

[1]The secondary diagnostic values and its definition are reported in `tables/table_p2f_summary.csv` and `reports/p2f_tighten.md`.

Table 5 External anchors vs. Graph-CoLD-soft under the identical P2f protocol: scenario-paired mean differences ($n = 30$ per dataset; positive = external method is better). All deltas shown are Holm-significant at $p < 0.05$ in the stated direction (Confident Learning above, Co-Teaching below Graph-CoLD-soft). Source: `tables/table_p5_external_tests.csv`.

Dataset	Method	ΔMacro-F1	ΔTail-F1	ΔTail-recall
CICIDS-2017	Confident-Learning	+0.048	+0.171	+0.189
	Co-Teaching	−0.155	−0.316	−0.201
CESNET-TLS-Year22	Confident-Learning	+0.114	+0.234	+0.283
	Co-Teaching	−0.204	−0.296	−0.163
UNSW-NB15	Confident-Learning	+0.014	+0.081	+0.089
	Co-Teaching	−0.066	−0.118	−0.109

External anchors under the identical protocol.

Table 5 places two external noisy-label methods, run under the identical P2f protocol (same audit windows, splits, seeds, and injected noise), against Graph-CoLD-soft with scenario-paired tests. The ordering is consistent across datasets. Confident Learning, a precise probabilistic pruner that removes only 7–24% of training rows, significantly outperforms Graph-CoLD-soft on aggregate and tail metrics on all three datasets. The signal-preserving weighting in turn outperforms hard deletion (Table 4). Full Co-Teaching collapses on tail classes (mean Tail-F1 ≈ 0 at 60–80% noise on every dataset; `tables/table_p5_tail_by_rate.csv`), because its small-loss forgetting systematically discards rare classes under this setup. This result is specific to the evaluated implementation and does not establish a general disadvantage of forgetting-based methods. Under the injected tail-to-majority noise model, precise pruning beats soft retention, because down-weighted flipped samples still teach the classifier that tail features indicate the majority target. We therefore scope the signal-retention claim to the mechanism level—within the shared pipeline, soft weighting recovers rare classes that hard deletion loses—and we identify Confident Learning as the performance bar that graph-based denoising must clear; combining confident-learning-precision pruning with graph-structured diagnostics is future work. Per-noise-rate tail-F1 breakdowns for all methods are reported in `tables/table_p5_tail_by_rate.csv`.

6.4 RQ3: A leakage audit prevents inflated claims

Our initial pipeline reported a $\sim$ 28-point CICIDS advantage with a curve that barely degraded under 60% noise—implausible for a denoiser. Auditing revealed that CICIDS-2017 contains 152,201/630,399 ($\approx$ 24%) exact-duplicate training rows, and that any clean-label signal leaking into the score/graph construction inflates results. We verified zero train/test edge crossings, removed exact and near-duplicate edges, and eliminated the clean-label paths (enforced by a regression test). The apparent advantage collapses to +6.7 points on CICIDS, +0.3 on CESNET, and a slight -1.3 on UNSW in the audited comparison (`reports/p2c_leakage_audit.md`). These audit deltas quantify the magnitude of the inflation and were measured under the earlier P2c audit protocol (a 3,000/8,000 stratified audit sample for CICIDS and the D5.5 frozen matrix for CESNET/UNSW); they aggregate differently from, and are not directly comparable to, the final P2f result matrix in Tables 2–4 (`reports/p2f_tighten.md`). We report this because graph denoising in security is unusually susceptible to such inflation.

6.5 RQ4: Graph informativeness is consistent with when the method helps

A per-dataset neighborhood-homophily / view-coverage diagnostic is directionally consistent with Graph-CoLD's margin: CICIDS (rich host/IP/temporal views) shows the largest gain, CESNET sits near a performance ceiling, and UNSW (temporal/behavioral only) shows the smallest—occasionally negative—margin (Figure 3, tables/table_p2d_graph_informativeness.csv). With only three datasets this pattern is descriptive, not inferential. This turns the UNSW result into an informative boundary condition rather than a failure: when views lack structural signal, the graph diagnostic adds little.

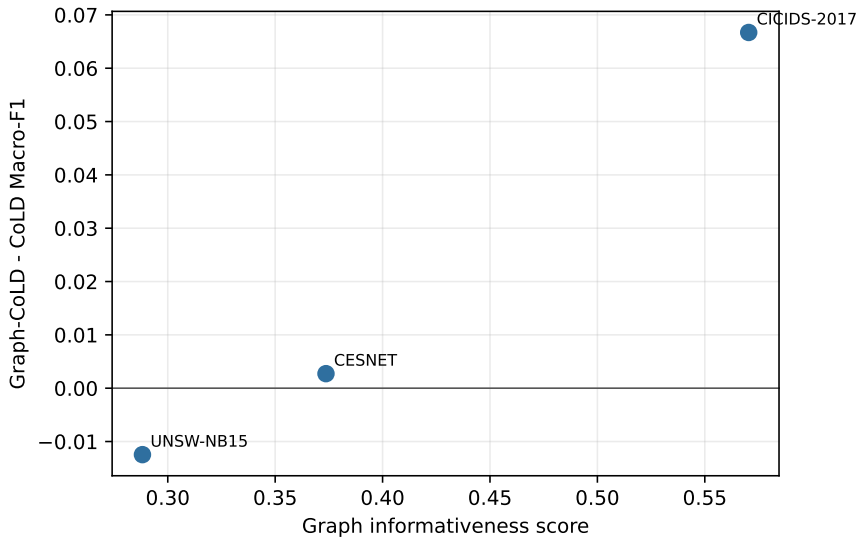


Fig. 3 Graph informativeness vs. Graph-CoLD margin over CoLD. Higher neighborhood homophily and view coverage are directionally consistent with larger gains (three datasets; descriptive pattern, not inferential).

6.6 RQ5: Cost

Graph-CoLD's overhead is moderate: mean runtime is comparable to CoLD and memory is lower than the strong purification baselines (the runtime and memory figures are measured under the earlier D5.5-protocol runs recorded in `results/runtime.json`; the relative pattern, not the absolute numbers, is the claim). Signal preservation adds negligible cost over hard deletion because it changes only the training weights.

7 Discussion

The operational motivation is strongest in settings where multiple telemetry sources provide partially consistent signals, such as host relationships, communication patterns, and temporal behavior. In those settings, a suspicious sample need not be useless: a rare attack instance may appear inconsistent because it represents an emerging technique rather than an incorrect label. Our benchmark study tests only this training-data mechanism; it does not establish deployment suitability.

However, the results also demonstrate clear boundaries. When available views are weak or incomplete, graph structure provides limited additional information. The UNSW-NB15 results illustrate this condition: without sufficiently informative relational views, graph-based diagnostics cannot reliably create additional training signal. Similarly, datasets operating near a performance ceiling, such as CESNET-TLS-Year22 under the evaluated protocol, naturally limit the observable improvement.

7.1 CoLD fidelity: what the proxy is and is not

The comparator historically labelled "CoLD" in earlier drafts is a *CoLD-aligned proxy*: a neighbourhood consistency diagnostic with hard thresholding and a shared classifier head, built before the official CoLD code release. It does not implement official CoLD's multi-subset feature encoding, clustering-consistency purification, or XGBoost pipeline. The official NDSS paper reports Macro-F1 in the 0.94–1.00 range on a nine-class refined CICIDS-2017 under random 8:2 splits—a protocol that differs from our 11-class, tail-targeted, group-disjoint audit windows in labelling space, split design, and noise model, so the numbers are not comparable in either direction. We use the proxy solely as a deletion-style anchor inside our frozen protocol and make no claim of reproducing or outperforming official CoLD [1]. An exact-alignment grid under the official protocol is registered as future work.

A key methodological decision in this work is the avoidance of unfair state-of-the-art comparisons. Two external anchors—Confident Learning and a full Co-Teaching implementation—were therefore run under the identical leakage-audited protocol and are reported in Table 5, even though one of them outperforms our method on every dataset; that result is part of the empirical record, not an inconvenience. Other purification and robust-training variants (MCRe, MORSE, FINE, Decoupling) were not all reimplemented under the final leakage-audited protocol. Mixing results obtained

under different data-cleaning, duplication-removal, and evaluation settings would create an unreliable comparison, so those methods remain excluded from the formal matrix. A comprehensive comparison with all previous methods requires faithful implementations or official code under a unified evaluation framework.

The external-anchor ordering carries a practical lesson for benchmark data curation and a hypothesis for later SOC-facing validation. Confident Learning's precise pruning—removing only 7–24% of rows—sets the performance bar under the injected noise model, while the evaluated Co-Teaching configuration nearly eliminates the tail classes. The differentiator is not deletion versus retention but *diagnostic precision*: an imprecise purifier can sacrifice scarce rare-class training signal, whether it deletes samples or down-weights mislabeled samples insufficiently. Combining confident-learning-precision pruning with graph-structured diagnostics is the natural next step.

The leakage audit provides a broader methodological lesson for graph-based security machine learning. Graph models can unintentionally exploit duplicated records, cross-split connections, or hidden clean-label information. Such effects may produce impressive but misleading improvements. Therefore, graph security studies should explicitly verify data separation, graph construction constraints, and label independence before reporting performance gains.

Future work should extend Graph-CoLD toward streaming SOC environments, full-archive evaluation, analyst-in-the-loop validation, and broader cross-domain benchmark comparisons. These extensions are necessary to determine whether signal-preserving denoising remains effective when labels arrive continuously and attack distributions evolve over time. The results support a bounded but useful claim. A label-space graph diagnostic modestly improves noisy-label IDS over CoLD where views are informative, and—more distinctively—signal-preserving weighting retains rare-class training examples that hard deletion discards, improving Tail-F1 under

high asymmetric noise within the shared pipeline. The current analysis detects no significant aggregate difference but does not establish equivalence. Under the same protocol, confident-learning pruning remains the stronger performer and marks the bar for future graph-based denoisers. We deliberately avoid three overclaims that our own experiments do not support: a universal accuracy advantage (the aggregate gain is modest and largely from the graph, not the weighting), a ranking advantage (Graph-CoLD ties strong peers on top-K precision), or SOC/forensic effectiveness from benchmark-level sample retention.

8 Threats to Validity

Internal. The evaluation uses deterministic audit windows for tractability; absolute Macro-F1 is therefore lower than full-archive runs, though the paired, seed-controlled comparisons are the object of inference. MCRe and MORSE pass a clean-label sanity floor but degrade on high-noise tabular settings more than in their original domains; we report this caveat rather than a possibly unfaithful number (`reports/p2b_baseline_fidelity.md`). *Construct.* The rare-evidence-recovery diagnostic, even after correction, saturates for soft retention; we therefore headline test-set tail-F1. *External.* CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 are useful benchmarks but not complete SOC deployments; the priority score is evaluated by proxy, not an analyst study. The two-dataset lock recorded in the D9 submission audit was an intermediate checkpoint; the authoritative final scope is the three-dataset P2f clean de-oracled protocol.

9 Limitations

MALTLS-22 and OpTC are not evaluated (no verified license/provenance path). Behavioral and threat-intelligence views are disabled where fields are absent. The aggregate accuracy contribution is modest and dataset-dependent; under the audited

protocol, Confident Learning outperforms Graph-CoLD on aggregate and tail metrics on all three datasets, so the distinctive contribution is the leakage audit, the mechanism-level signal-retention finding, and the informativeness boundary analysis rather than state-of-the-art accuracy. Full-archive and streaming evaluations are left to future work.

10 Conclusion

We revisited collaborative label denoising and asked whether a multi-view label-space graph diagnostic and signal-preserving weighting improve robustness. The answer is targeted: the graph diagnostic gives a modest, informativeness-dependent accuracy gain over CoLD, and signal-preserving weighting improves Tail-F1 over hard deletion under high asymmetric noise within the shared pipeline. The current aggregate analysis detects no significant difference but does not establish equivalence. A confident-learning anchor, evaluated under the identical protocol, outperforms both. Alongside, we contribute a leakage audit and a graph-informativeness diagnostic that together explain when—and when not—graph denoising helps. More broadly, our results argue that security data curation should shift from indiscriminate purification toward signal-aware retention, in which deletion decisions explicitly weigh the class-level cost of discarding scarce training examples against the precision of the noise diagnostic. CESNET supports only a service-class stress-test claim; attack-level conclusions are limited to CICIDS and UNSW. We hope the audit, and the audited external-baseline ordering, are as useful to the community as the method.

Appendix A CESNET-TLS-Year22 class-filter policy

CESNET is evaluated as a TLS service-classification stress test, not as an attack dataset. The repository audit report `reports/cesnet_class_policy_report.json`

records 178 raw service labels. The frozen post-filter policy retains the 25 labels in Table A1 because each has at least 1,000 rows in the audited source window; the other 153 labels are excluded rather than merged. The largest retained class is capped at 4,958 rows. Counts below are pre-split source-window counts and are not labels of attack prevalence.

Table A1 Frozen CESNET 178-to-25 service-class filter.

Service label	Count	Service label	Count
apple-itunes	1,666	bitdefender-gravityzone	1,943
dropbox	2,472	eset-edf	1,506
eset-edtd	1,324	facebook-graph	1,989
google-www	1,088	instagram	1,323
kaspersky	1,459	microsoft-authentication	3,083
microsoft-defender	1,326	microsoft-diagnostic	4,958
microsoft-onedrive	3,137	microsoft-push	3,516
microsoft-settings	4,808	microsoft-update	4,958
o2tv	1,989	office365	2,584
outlook	3,517	seznam-media	1,787
skype	3,511	spotify	1,217
teams	3,802	tiktok	1,128
vmware-vcsa	1,033		

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